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Prediction of the Tandem Propeller Performance Using Linear Superposition Method

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We continue our study on the aerodynamic investigation of two propellers mounted partially overlapping and in tandem configurations. In the previous study, the force-based experiment was conducted in both positive and negative staggered configurations. The current study implements smoke flow visualization and Particle Image Velocimetry (PIV) techniques to further investigate the propeller flowfield in tandem configuration. All experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open-jet configuration. A decrement in back propeller inflow angle was observed in the flow visualization experiment and confirmed by the PIV measurement. By applying the propeller performance prediction model developed in the previous study, the changes in the back propeller thrust generation in tandem configuration were predicted.

I. Nomenclature

α_{in}	=	Propeller Inflow Angle, (deg)
C_T	=	Propeller Thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$
C_P	=	Power Coefficient; $C_P = 2\pi\, C_Q = \frac{P}{\rho n^3 D^5}$
C_Q	=	Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$
D	=	Propeller Diameter, (m)
d	=	Distance Between Propeller Hubs, (m)
J	=	Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$
n	=	Propeller Rotational Speed, (RPS)
P	=	Power, (W)
γ	=	Propeller Pitch, (m)
Q	=	Torque, (Nm)
R	=	Propeller Radius, (m)
T	=	Thrust, (N)
u	=	Flow Velocity on X-axis, (m/s)
v	=	Flow Velocity on Y-axis, (m/s)
V_∞	=	Freestream Velocity, (m/s)
ρ	=	Air Density, (kg/m^3)
ψ	=	Propeller Phase Angle, (deg)
θ	=	Propeller Incidence Angle, (deg)
$\theta_{v\infty}$	=	Freestream Flow Angle, (deg)

II. Introduction

The increasing usage of propeller-driven aircraft, including unmanned aerial vehicles (UAVs) and eVTOL air taxis, broadens the classical helicopter problem of edgewise flight. One common method to increase the payload of the eVTOL vehicle is to increase the number of propellers, which also helps achieve a lower disk loading, thereby increasing efficiency and range [1]. However, eVTOL vehicles commonly face the constraint of minimizing their overall size or ground footprint to facilitate operations in small spaces. Coaxial rotors are considered a solution for reducing the footprint, a concept occasionally employed in helicopters to minimize torque imbalance [1]. However, an average performance loss of 15 ~ 25% was reported when compared to a single

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rotor in hover, as the lower rotor operates constantly in the wake of the upper rotor [2-5]. This loss is reduced to 9 ~ 15% in high-speed forward flight [2,6,7].

An alternative to a coaxial rotor is to mount the propellers in close proximity within the same plane similar to the Volocoptor [8]. The loss in hover performance is negligible when the propellers are placed over 1.1 d/D apart, between rotor hubs [5, 10–12, 17]. However, a performance loss of 11 ~ 18% in low-speed forward flight is observed when $d/D < 3$, due to rotor interaction in forward flight adversely affecting the downstream rotor [2, 6, 7, 11-18]. Moreover, an overall thrust difference between the front and rear rotor can be up to ~40% based on flight test data in Ref. [13] and Ref. [16-17]. To minimize the rotor-to-rotor interference, the rotor can be placed in a tandem configuration with a vertical displacement. Based on flow visualization studies and force-based testing, Ref. [13] suggests that the reduction of the back rotor's performance is due to the changes in its inflow angle caused by the front rotor, which agrees with the Fast Multirotor Performance Prediction Model (FMPP) developed by Tsaltas [14] and the panel method analysis conducted by Kostek et.al. [18].

In our previous study [21,22], the performance of a tandem overlapping propeller pair was investigated with the rear propeller placed in both negative (back propeller above the front propeller) and positive tandem (back propeller beneath the front propeller) configurations. For the negative tandem, increasing the vertical spacing between the propellers results in a decrease in power required at constant thrust at an advanced ratio $J > 0.2$, or in other words, an increase in efficiency in forward flight. On the other hand, a significant reduction in the back propeller performance was measured in positive tandem configuration due to the direct wake impact front propeller. The reduction in back propeller performance was also sensitive to the advance ratio and the vertical spacing. The current study aims to further understand the cause of the changes in back propeller performance by investigating the propeller flowfield. Moreover, a prediction model is proposed to estimate the back propeller performance in tandem configuration using the linear superposition principle, which provides a fast performance estimation method during the initial multi-propeller air-vehicle design stage.

III. Experimental Setup

A) General Experimental Setup

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test section. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The turbulence intensity of the testing section is 0.1% measured by a hot wire anemometer. A sketch of the experimental setup is shown in Figure 1. The KDE CF-125 [23] propeller is paired with a KDE 2814-XF-515 brushless motor and Hobbywing Platinum V4 electronic speed controller operating in governor mode for better rotational speed control. The propeller and motor assembly are mounted on a 3D-printed interface, bolted to an 80-20 aluminum extrusion frame on a Fuyu FSL 40 linear traverse. The linear traverse is placed at the centerline of the test section. It changes the vertical distance (z) between the propellers. The propellers can be tilted with respect to the tunnel freestream. To change the downstream distance (y), the location of the front rotor is fixed, while the rear propeller is moved back or forth. The entire assembly is mounted to a 1.22 x 0.92 x 2.44 meter aluminum frame, secured to the lab floor, the inlet of the wind tunnel, and the ceiling of the wind tunnel plenum.

B) Flow Visualization

Smoke flow visualization was performed with the front propeller rotating at 67 RPS. A smoke wand, with a diameter of 0.5 inches, was placed at 0.5 D upstream of the propeller at the inlet of the testing section. A 300-mW, 532 nm continuous laser was used as a light source. All images are taken by a Panasonic LUMIX GH5 digital camera with a Cannon EF 50 mm f/1.4 lens. The advance ratio of the back propeller for smoke flow visualization was between 0.1 and 0.5.

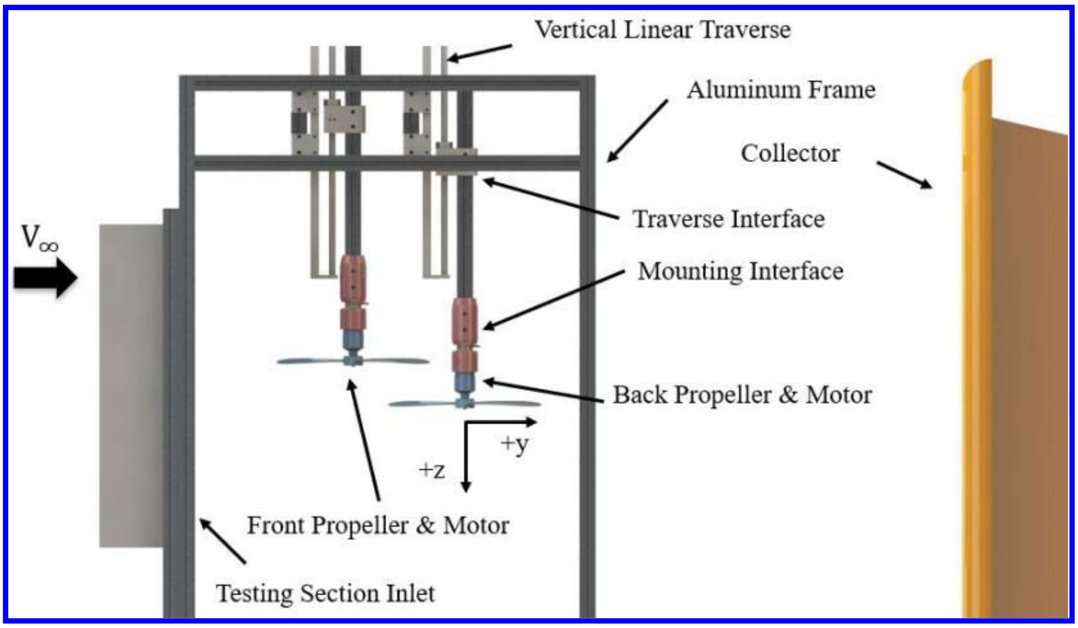

Figure 1. Schematic of the experimental setup at the UD-LSWT

C) Particle Image Velocimetry

Streamwise, non-phase-locked, and phase-locked Particle Image Velocimetry (PIV) was conducted to investigate the propeller flowfield for the standalone propeller and the back propeller in tandem propeller configuration in forward flight. The tracer particles for the experiments were generated by a ViCount smoke seeder with glycerin oil. The Quantel Twins CFR 300, 200 Mj/pulse Nd: YAG (neodymium-doped yttrium aluminum garnet) frequency-doubled laser was aligned with a 60-degree fan angle Powell lens to illuminate the flow field. The Imperx 2021 CCD camera was used with a Canon EF 100mm f/1.4 lens to provide a field of view of 127 mm x 127 mm for all cases. To trigger the camera shutter during the phase-locked PIV experiment, a 50 mW, 532 nm laser, and a photodiode were used to trigger the Quantum Composers 9600 pulse generator, which captured the flow passing different propeller phases by changing the delay on the QC pulse generator.

For all cases tested, 1000 image pairs were obtained per case and processed using the Fluere software (Version 1.3). Two iterations were performed in processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. The time delay between the laser pulses was changed at each advance ratio to obtain 8 to 12-pixel particle displacement at the region of interest in the flowfield. According to the PIV uncertainty estimation developed by Lazar et al [58], which includes uncertainty from equipment (instrumentation), particle dynamics, sampling, and image analysis (processing), the average uncertainty in velocity is around $\pm 3.59\%$ for all cases tested.

D) Test Matrix

Two different PIV campaigns are conducted in the current study. The investigation of the standalone propeller focuses on the propeller inflow as well as its downstream wake, while the investigation of the tandem propeller focuses only on the back propeller inflow.

D. 1 Standalone Propeller Investigation

The PIV test matrix for the standalone propeller is shown in Table 1. The schematic of the field of view (FOV) location for the experiment is shown in Figure 2, all field of view has a size of 127 mm x 127 mm. For the investigation of the standalone propeller inflow, FOV A is used which approximately covers the propeller blade from 0.35 r/R to the tip of the propeller. The propeller rotational speed is fixed at 100 RPS and four different incidence angles (θ) are investigated. All images at FOV A are phase-locked to better compare with the PIV result from the tandem propeller cases. On the other hand, FOV B to F is used when investigating the propeller downstream flowfield induced by the propeller. A 25.4mm overlap of each FOV is selected to better reconstruct the flowfield. The phase-locked technique is not used when investigating the propeller downstream flowfield as the front propeller in the tandem propeller PIV campaign is not phase-locked. The rotational speed of the propeller is set at 67 RPS to match the front propeller rotational speed in the tandem configurations. Only one incidence angle is tested. For both FOV locations, five different advance ratios (J) are tested with 1000 image pairs for each case.

Table 1. PIV Test Matrix for Standalone Propeller Experiment

Propeller	Field of View Location	Advanced Ratio	Rotational Speed (RPS)	Incidence Angle (Deg)	Phase Locked	Image Pairs
KDE 12.5x4.3	A	0.1, 0.2, 0.3, 0.4, 0.5	100	45, 60, 75, 90	Yes	1000
	B - F		67	90	No	

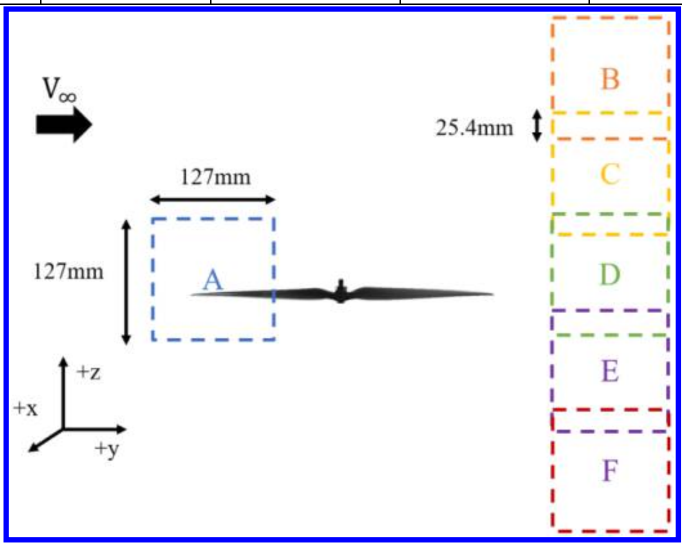

Figure 2. Schematic of Standalone Propeller PIV Field of Views

D.2 Tandem Propeller Investigation

The test matrix for the tandem propeller experiment is shown in Table 2, and the schematic of the FOV is shown in Figure 3. Only one FOV is selected for the back propeller, which is identical to the FOV A of the standalone propeller in Figure 2. To achieve the best comparison of results, the camera and back propeller location are fixed, while the front propeller location varies. Five advance ratios are tested with the front propeller rotational speed fixed at 67 RPS while the back propeller rotational speed varies to maintain the trim condition based on the results from the force-based experiment. 8 different propeller vertical placements are investigated with the back propeller placing 1.25 d_y/D downstream from the front propeller.

Table 2. PIV Test Matrix for Tandem Propeller Experiment

Propeller	Field of View Location	Advanced Ratio	Rotational Speed (RPS)	Incidence Angle (Deg)	Vertical Placement d_z/D	Downstream Placement d_y/D

KDE 12.5x4.3	A	0.1, 0.2, 0.3, 0.4, 0.5	Front: 67 Back: 67 - 105	90	-0.8~0.6	1.25

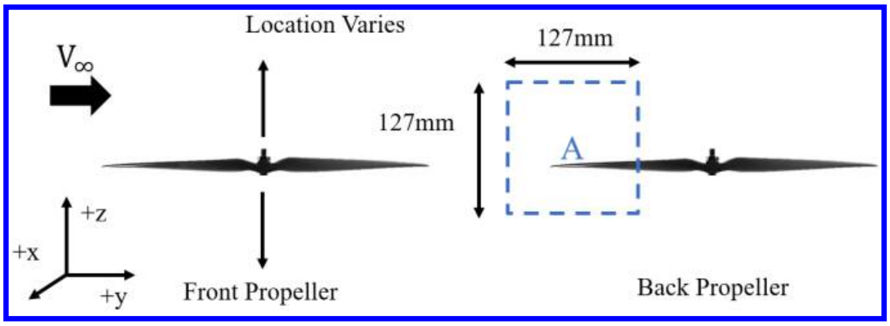

Figure 3. Schematic of Tandem Propeller PIV Field of View

IV. Results

For all flow visualization and PIV experiments, the tandem propeller is operating under trimmed conditions, which is consistence with our previous study [21,22]. To trim the tandem propeller pair in cumulative pitching moment, the rear propeller rotational speed is adjusted such that $C_{Qx\ norm}$, less than 5% for each advance ratio. This is shown in Figure 4 and $C_{Qx\ norm}$ is defined as:

$$C_{Qx\ norm} = \frac{C_{Qx\ Front} + C_{Qx\ Back} + (T_{Front} - T_{Back}) * 0.5d_Y}{C_{Qx\ Front} + C_{Qx\ Back}} \approx 0 \tag{1}$$

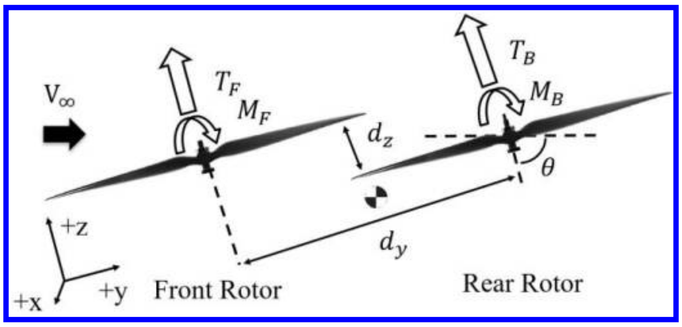

Figure 4 Nomenclature for tandem propellers.

A) Flow Visualization

Smoke visualization is then conducted to investigate the flowfield of the tandem propeller system in forward flight. The $d_y = 1.25$ cases are selected for all smoke flow visualization cases as the effect of horizontal propeller spacing is relatively small. The back propeller inflow is first investigated, the inflow angle is defined as the angle between the propeller inflow and the propeller axial direction, as denoted as α_{in} in Figure 5a. Three propellers vertical displacement was selected: $d_z = 0.4$ (the best performer), $d_z = 0.0$ (traditional orientation) and $d_z = -0.4$ (worst performer).

At $J = 0.1$, the inflow angle for all three cases is almost identical. This is also reflected in the force-based experiment result where the difference in the tandem propeller system performance is almost negligible. Increasing J to 0.2, the differences in α_{in} begin to occur. It is also worth noting that the wake of the front propeller for the $d_z = -0.4$ case impacts directly on the rear propeller. This is also reflected in the results of P_t/P_s in our previous study [22] where a sudden spike in power consumption for the $d_z = -0.4$ case occurs at

J~0.2. This trend preserves at $J = 0.3$. For the highest J tested, the difference in α_{in} between $d_z = 0.4$ and $d_z = 0.0$ increases further. This agrees with the result in our previous study [21,22] where the overall performance difference between these two cases also reaches its maximum at $J \geq 0.4$. On the other hand, although the α_{in} difference between the $d_z = 0.0$ and $d_z = -0.4$ is small, and the rear propeller for the $d_z = -0.4$ continuously operating in the wake of its front propeller. As the inflow angle reduces while the inflow velocity increases, the thrust generated on the back propeller reduces. This leads to a lower overall tandem system performance for the $d_z = -0.4$ case. In general, a higher α_{in} leads to a better performance of the rear propeller and, hence, a better overall performance of the tandem propeller system.

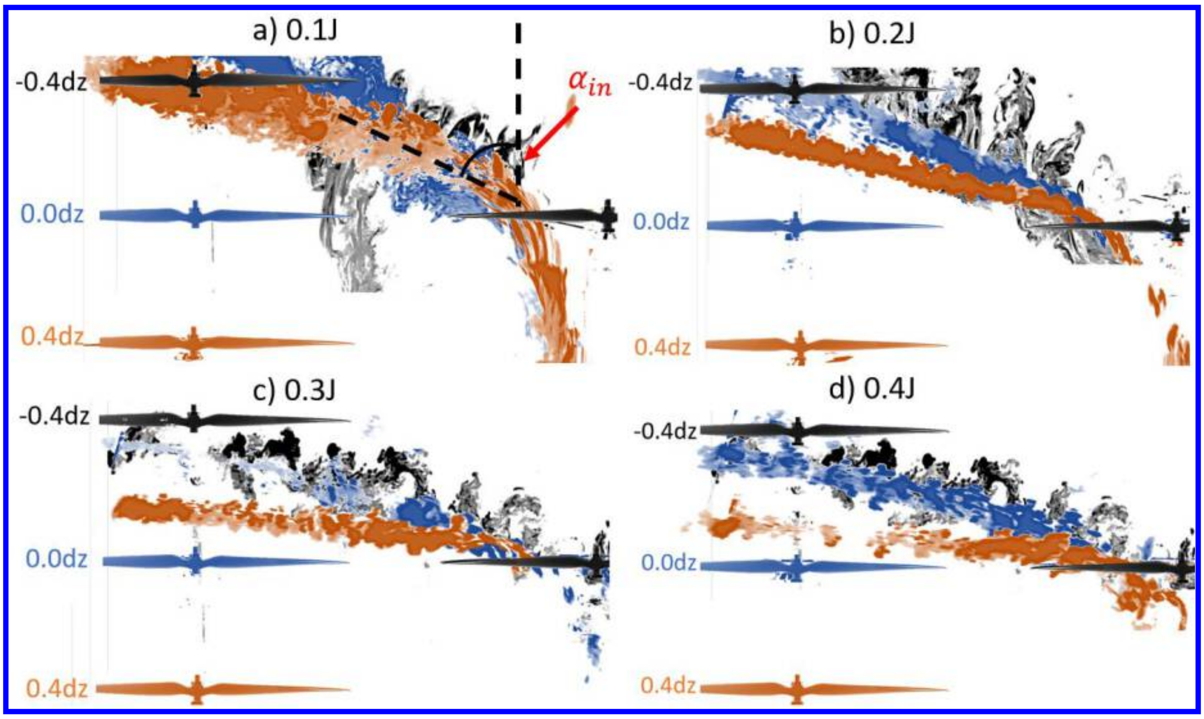

Figure 5. Smoke Flow Visualization Results for $d_z = -0.4, 0.0, 0.4$ at a) 0.1J b) 0.2 J c) $0.3J$ and d) 0.4J and $\theta = 90°$

B) PIV Results for Standalone Propeller Inflow at Different Incidence Angles

To better understand the effect of the propeller inflow angle on its performance in forward flight, the PIV investigation is first conducted to investigate the flowfield around the standalone propeller. As shown in Figure 2, one field of view (FOV) close to the propeller 0-degree phase angle (ψ) is chosen to investigate the propeller inflow, and five FOVs downstream of the propeller are selected to investigate the propeller-induced velocity and angle in the flowfield. This will also be used in the linear superposition modeling in the next section.

For all PIV investigations of propeller inflow, phase-locked PIV is conducted with the image taken at $\psi = 0°$ which minimizes the laser reflection in the field of view as well as avoids the interference of the propeller blade in the PIV image. While the non-phase-locked PIV is conducted at the propeller downstream as the front and back propellers in the tandem configuration do not operate at the same rotational speed. Figure 6 below shows an example of all FOVs for the standalone propeller with respect to the propeller locations. The contour in Figure 6 represents the freestream inflow angle, $\theta_{v\infty}$, which is calculated based on the propeller orientation at $\theta = 90$ for better results comparison across different cases. Therefore, $\theta_{v\infty} < 90°$ indicates a downwash in the flowfield while $\theta_{v\infty} > 90°$ indicates an upwash in the flowfield. For each θ condition, the propeller inflow angle can be calculated as:

$$\alpha_{in} = \theta_{v\infty} + (\theta - 90°) = \text{atan}\left(\frac{V_x}{V_y}\right) + (\theta - 90°) \tag{2}$$

Where $\alpha_{in} = 0°$ indicates that the propeller inflow is parallel to the propeller axial axis and $\alpha_{in} = 90°$ indicates the propeller inflow is perpendicular to the propeller axial axis. The results at FOV at the propeller inflow will first be discussed, and the results at FOVs downstream of the propeller will then be discussed as a group.

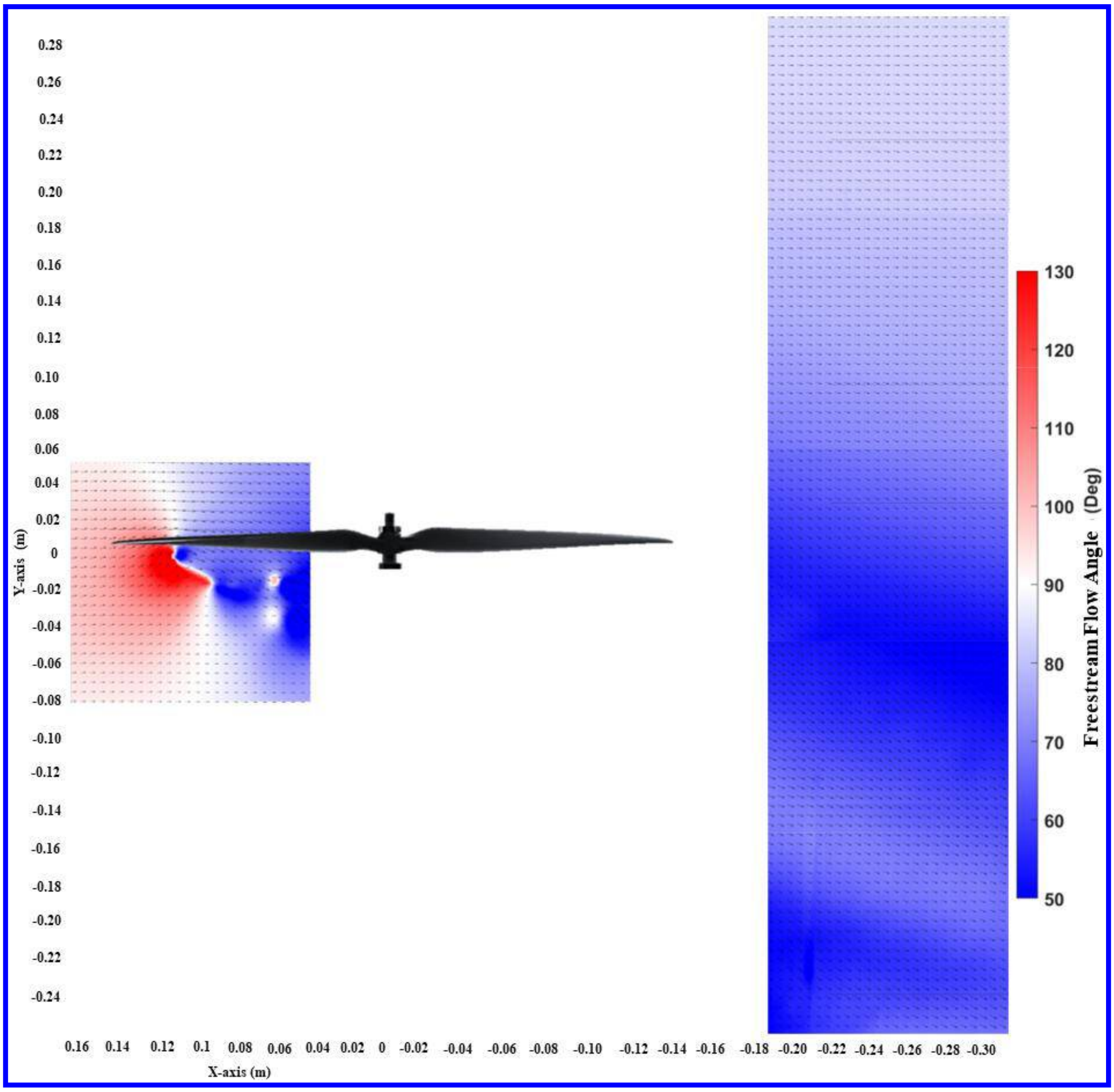

Figure 6. Example PIV Showing all FOVs for Standalone Propeller at $J = 0.3$ and $\theta = 90°$

B.1 Standalone Propeller Inflow

Figure 7 shows the PIV results for the standalone propeller at $\theta = 60°, 75°$ and $90°$ under various J. The annotated propeller blade in all PIV images indicates the propeller disk location, while it does not represent the actual propeller location when the PIV image is taken. The propeller inflow is shown in the form of streamline and vorticity contour. For all cases, α_{in} (Equation 2) is calculated as half propeller mean aerodynamic chord upstream from the propeller disk, which is also indicated as the red dashed line in Figure 7. Meanwhile, the measured vorticity in the PIV image is normalized by the propeller diameter and the propeller blade speed at the 75% radius location, which can be written as:

$$\omega_{norm} = \frac{\omega * D}{n * \pi D * 0.75} \tag{3}$$

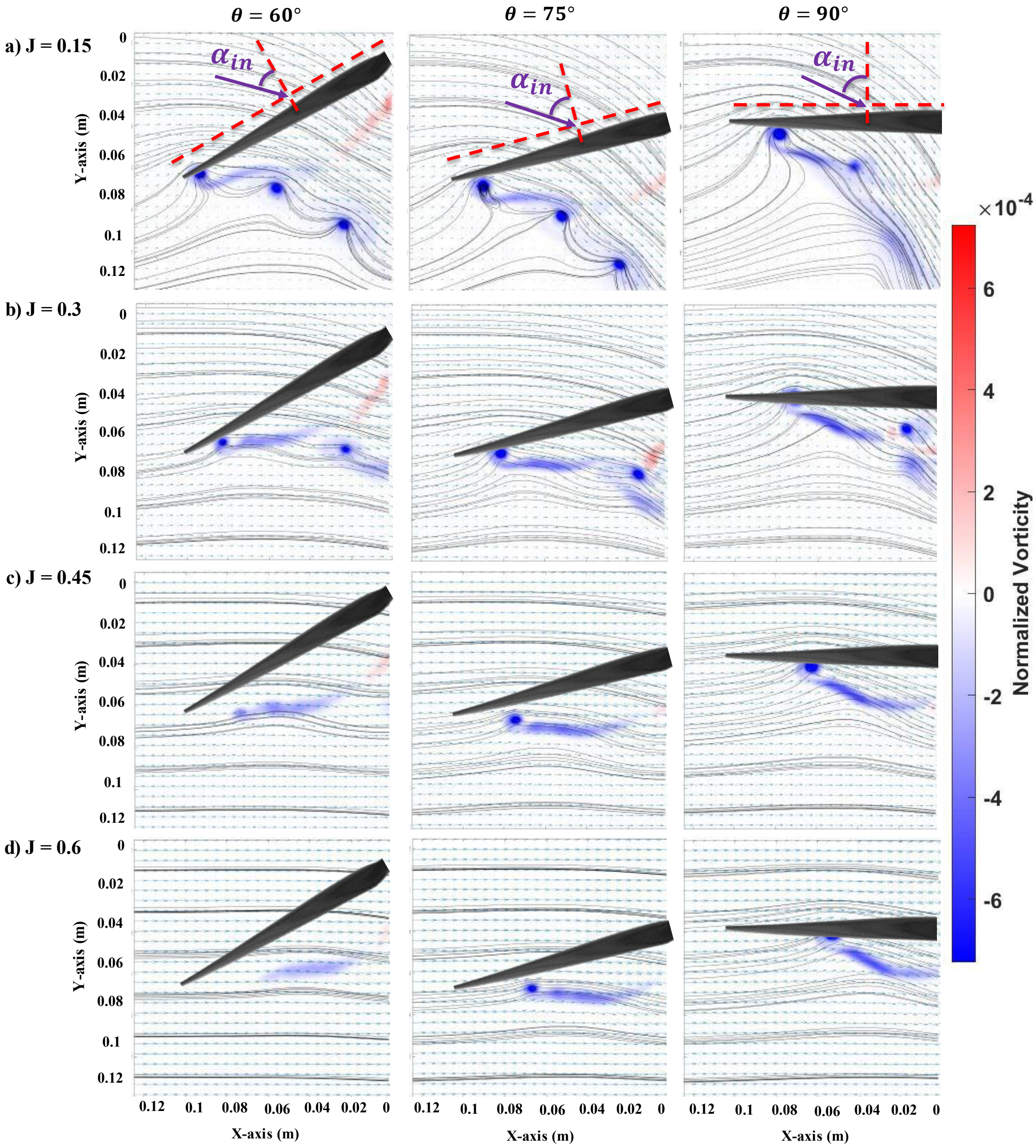

Figure 7. PIV Results for Standalone Propeller Inflow at $\theta = 60°, 75°\ 90°$ Under a) $J = 0.15$, b) $J = 0.3$ c) $J = 0.45$ and d) $J = 0.6$

It is obvious that the propeller momentum dominates the flowfield at $J = 0.15$ where the value of α_{in} for all θ cases are relatively low and closely resembles the hovering condition. A strong upwash is also observed closer to the tip of the propeller disk, as a result of the propeller tip vortex rollup. With the increment of J to 0.3, the freestream momentum increases while the strength of this tip upwash as well as the downwash above the propeller disk reduces for all cases, resulting in the calculated freestream flow angle trending towards $90°$. This trend is most noticeable for the $\theta = 60°$ case where the thrust generated by the propeller is the lowest among these three cases, and therefore the momentum added to the freestream is also the lowest based on the momentum theory. This trend persists with further increment in J. At the highest J tested, the measured $\theta_{v\infty}$ is almost uniform at the propeller disk with the calculated value of $\sim 90°$ for the $\theta = 60°$ case, while the upwash can still be observed for the $\theta = 75°$ and $90°$ cases, with a higher θ showing a higher influence on the freestream flow angle.

The calculated α_{in} for each case is shown in Figure 8 for all cases shown in Figure 7 and an additional $\theta = 45°$ case. A linear α_{in} distribution, and thus a linear downwash distribution is observed for all cases at $r/R < 0.75$, while a sudden increment in α_{in} is observed closer to the tip of the propeller blade as a result of the upwash induced by the tip vortex. This indicates that the traditional linear inflow model for helicopter rotors [1] is relatively accurate and can be applied to propellers for simple modeling purposes, however, the actual inflow distribution is highly non-linear closer to the tip of the rotor and difficult to model without experimental data [25].

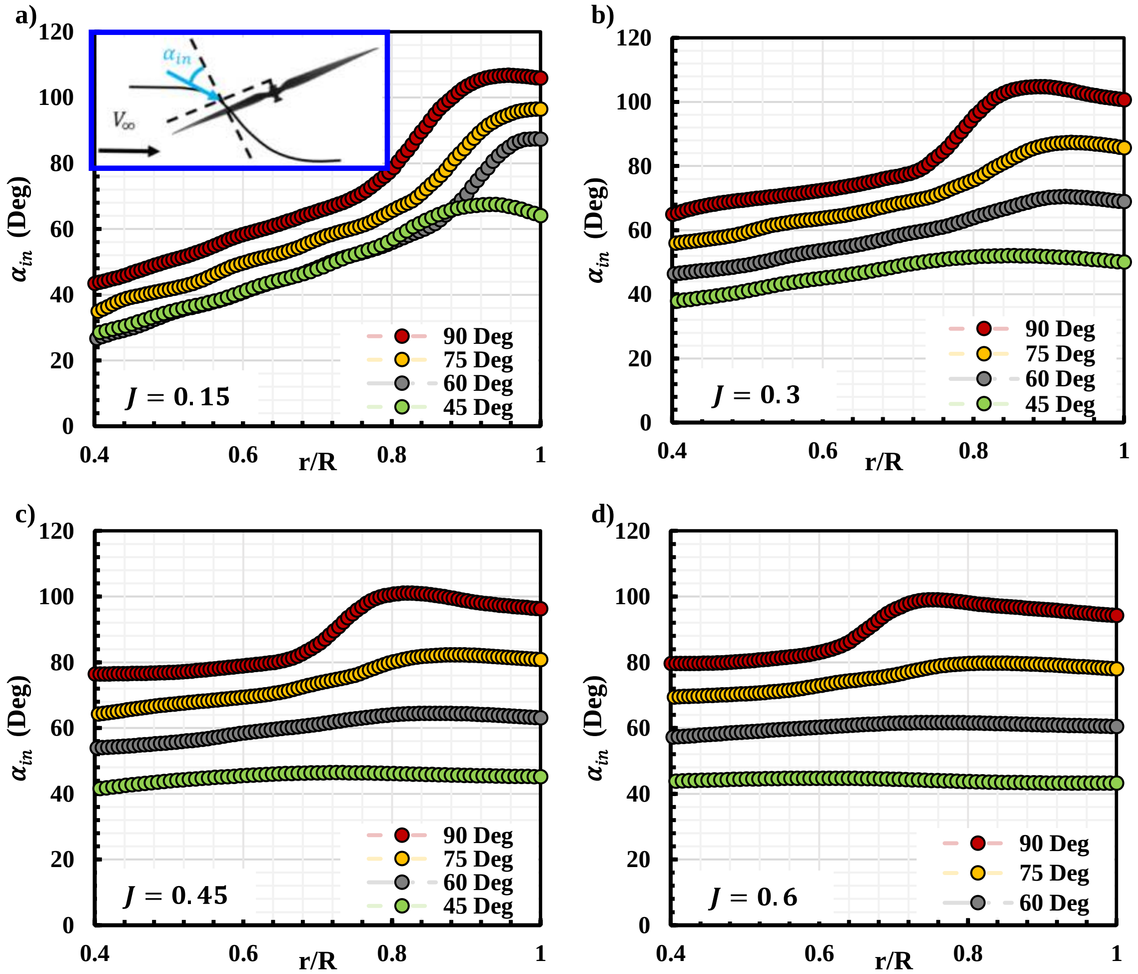

Figure 8. Standalone Propeller α_{in} Distribution at Different θ and a) $J = 0.15$, b) $J = 0.3$ c) $J = 0.45$ and d) $J = 0.6$

With the increment in J, the difference in α_{in} the distribution between each θ case also increases. Further increment in J leads to α_{in} distribution reaching towards θ at a large, as a result of the increment in freestream momentum. For $\theta = 45°$ and $60°$ cases, α_{in} is almost equal to θ at $J = 0.6$, indicating that the propeller is no longer inducing a velocity at the propeller disk and adding momentum in the freestream, which agrees with the force-based measurement [21,22] where no thrust is generated at $\theta = 45°$ and a very small C_{Tz} is measured at $\theta = 60°$. It is also worth noting that the slope of the α_{in} distribution, $d\alpha_{in}/d\,(r/R)$ also reduces with the increment of J due to the increment of freestream momentum. For $\theta = 75°$ and $90°$ cases where the measured C_{Tz} is equal or greater than C_{Tz} at hovering, the upwash of the freestream is still observed at the propeller blade tip while $\alpha_{in} < \theta$ at $r/R < 0.75$, as a result of the lift/thrust generation on the propeller blade.

In order to systematically quantify the changes of α_{in} with respect to θ, the propeller inflow angle at a 50% radius, which avoids the upwash region closer to the propeller tip due to the tip vortex, represents the overall propeller inflow at each incidence angle. The results are shown in Figure 9, along with the theoretical calculation based on the BET methods [1] using the KDE propeller geometry information [23]. At a low advance ratio, the measured α_{in} is higher than the calculated value. This difference is reduced with the increment of J and matches with the BET calculation relatively well at $J > 0.3$. Therefore, the experimental data can be used to create a three-dimensional interpolation surface between α_{in}, θ and J for future modeling.

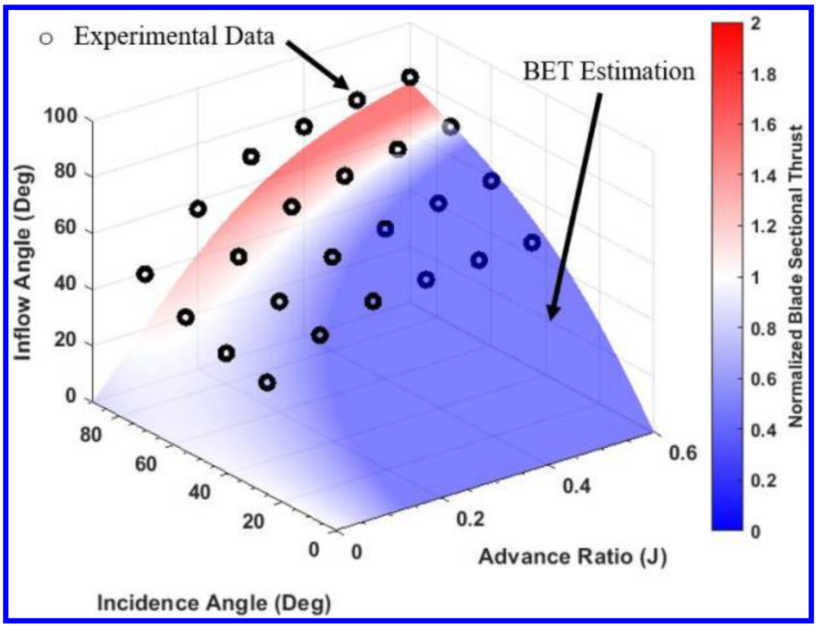

Figure 9. Measured (Scatter Data) KDE Propeller Inflow Angle at $0.5r/R$ and $\psi = 0°$ at Various J and θ Compared to Calculation from BET Method (Surface)

B.2 Standalone Propeller Wake

According to the classical momentum theory, the propeller would induce an additional velocity when thrust is generated. Therefore, for a propeller operating at a large incidence angle, the induced velocity vector will be almost perpendicular to the freestream, resulting in a downwash in the propeller downstream flowfield both above and below the propeller disk, and thus reducing $\theta_{V\infty}$ of the freestream. In this case, the downstream propeller would experience a lower α_{in} and therefore a performance loss, which is observed in the force-based experiment in our previous study [21,22]. The non-phase-locked PIV results conducted at the KDE 12.5x4.3 propeller downstream are shown in Figure 10 for all five FOVs combined together at $J = 0.1, 0.2$ and 0.3. The annotated red dash line represents the back propeller location in the tandem configuration at $d_y = 1.25$. However, none of the back propellers were actually placed in the FOV during the experiment.

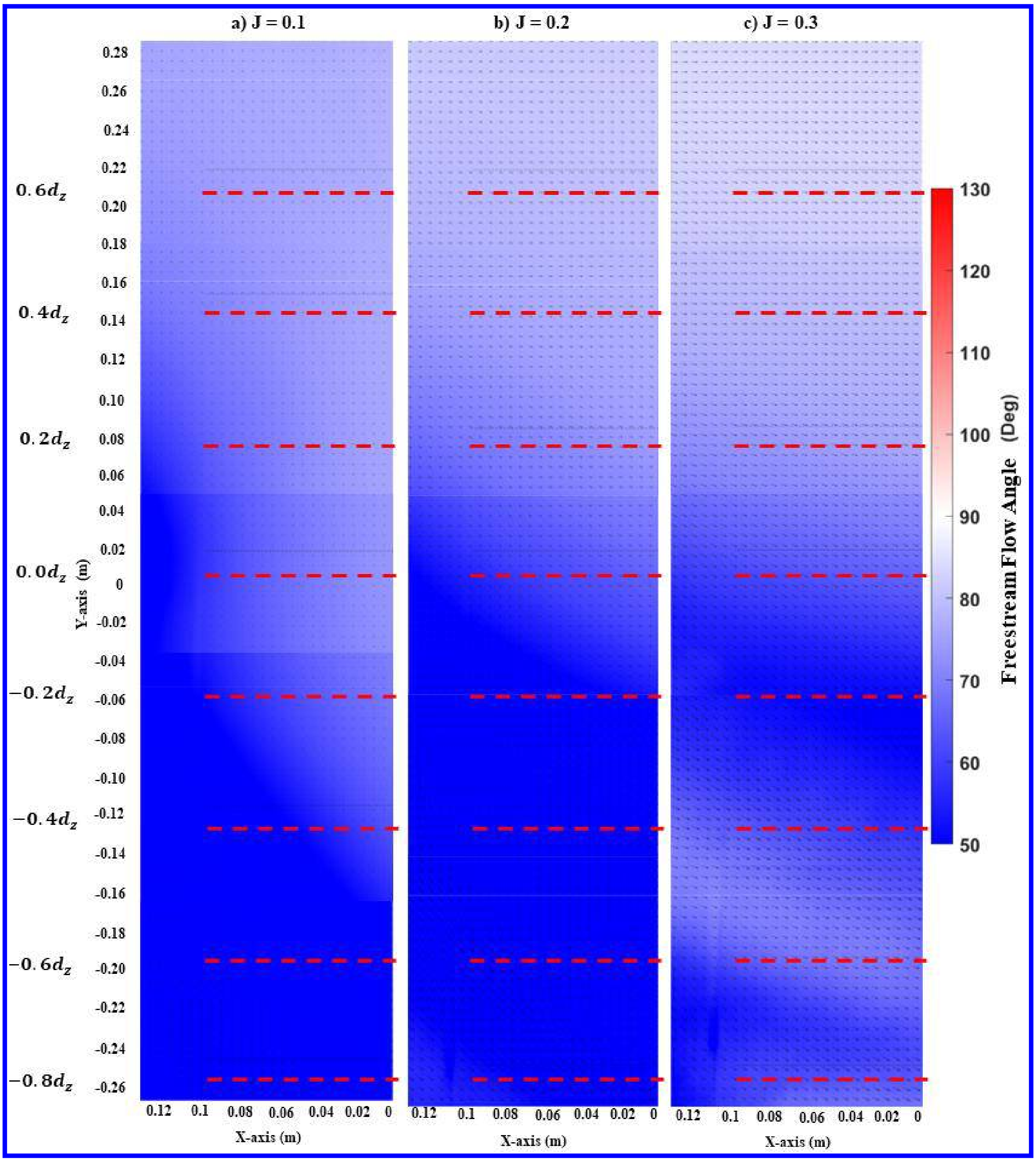

Figure 10. PIV Results for Standalone Propeller Downstream Flowfield at $\theta = 90°$ a) $J = 0.1$, b) $J = 0.2$ and c) $J = 0$.

At $J = 0.1$, a strong reduction in $\theta_{V\infty}$ is observed at the back propeller location of $d_z < 0.4$, where the wake of the front propeller is observed. At higher vertical locations, although a downwash is observed, the overall velocity vector is small, indicating a relatively small momentum of the flow in this area. Increasing J to 0.2, the wake of the front propeller starts to affect more back propeller locations as a result of the increment in the front propeller wake angle. At $d_z > 0.2$, $\theta_{V\infty}$ also increases when compared to the $J = 0.1$ case due to the increment in the freestream momentum. This trend is preserved at higher advanced ratios as well.

Figure 11 shows the calculated back propeller-induced inflow angle ($\alpha_{in\ ind}$) distribution estimation at different d_z locations due to the front propeller interference at various advanced ratios. Again, the back propeller is not placed in the flowfield. At $J = 0.1$, a clear order of the $\alpha_{in\ ind}$ magnitude is observed, with the highest d_z cases seeing the highest measured $\alpha_{in\ ind}$, indicating a smaller influence by the front propeller. When decreasing d_z, the overall measured $\alpha_{in\ ind}$ reduces, with the $d_z = 0.8$ case seeing an average $\alpha_{in\ ind}$ of ~15°, as the wake of the front propeller is directly impacted this location.

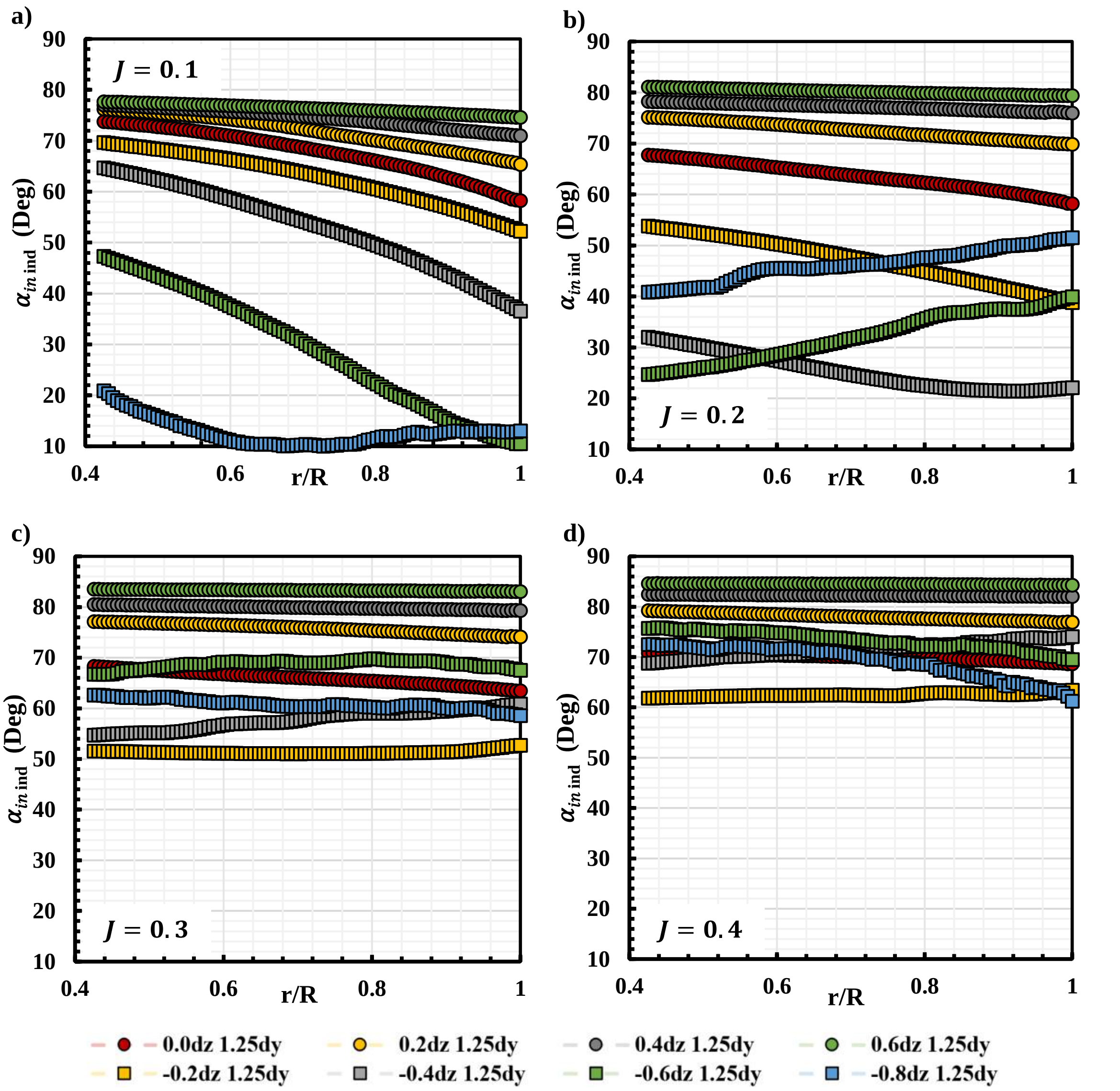

Figure 11. Measured Induced Back Propeller Inflow Distribution due to Front Propeller at a) $J = 0.1$ b) $J = 0.2$ c) $J = 0.3$ and d) $J = 0.4$.

Increasing J to 0.2, the order of $\alpha_{in\ ind}$ magnitude remains the same for $d_z \geq 0.0$. Magnitude of $\alpha_{in\ ind}$ for $d_z = -0.8$ increases as a result of the increment in the front propeller wake angle, resulting in the minimum measured $\alpha_{in\ ind}$ at $d_z = -0.4$. The front propeller wake angle continues to increase at higher J resulting in the minimum $\alpha_{in\ ind}$ at $d_z = -0.2$, while the overall order of $\alpha_{in\ ind}$ the magnitude for other cases remains the same as previous, this trend continues at higher J where the minimum $\alpha_{in\ ind}$ is measured at $d_z = -0.2$ due to the continuous direct wake impact on this location.

Using a similar method to quantify the propeller inflow angle, α_{in}, the downstream induced flow angle due to the front propeller can also be quantified at the $0.5\ r/R$ location of the back propeller. The results are shown in Figure 12 below along with the calculated front propeller far wake skew angle, χ_{prop} from the BET as contour color, which is defined as:

$$\chi_{Prop} = \text{atan}\left(\frac{V_\infty\ sin\theta}{V_\infty\ cos\theta + 2V_i}\right) \quad (4)$$

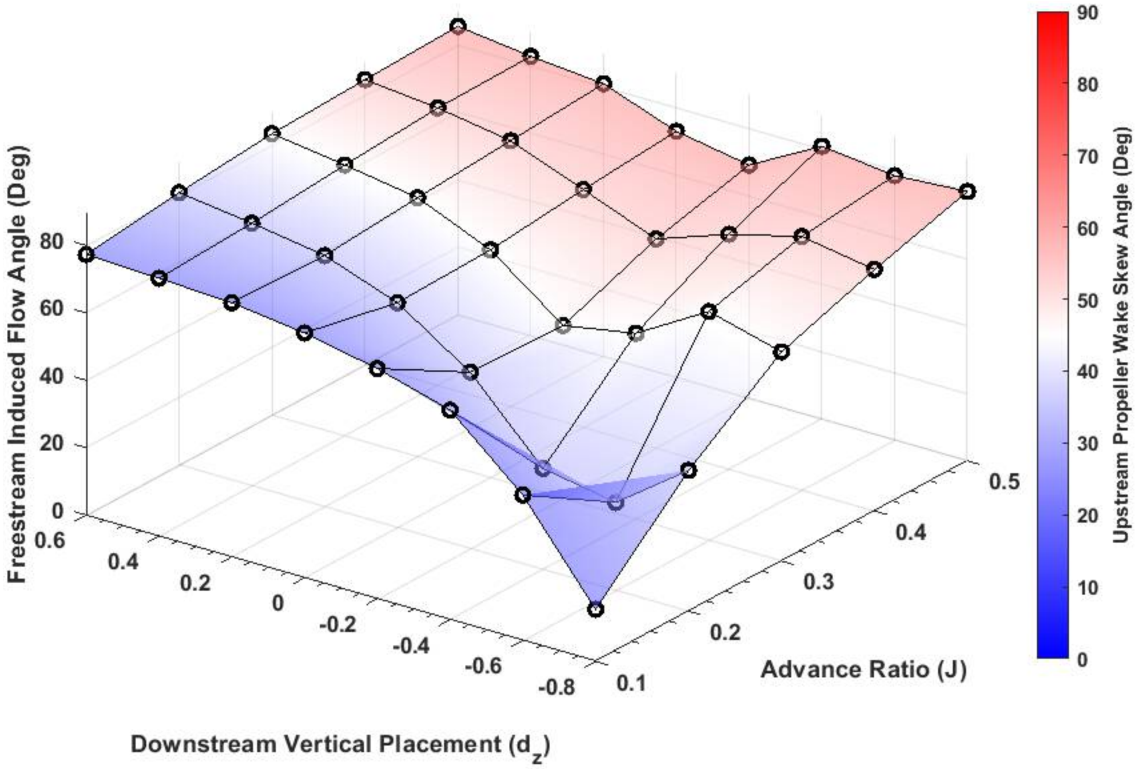

Figure 12. Downstream Induced Flow Angle due to Front Propeller at Various J and d_z.

With the relationship between the propeller thrust generation, propeller incidence angle, forward speed, as well as the inflow angle determined, the propeller performance due to various interactions can be modeled by quantifying the changes in its effective inflow angle and estimating the effective incidence angle at a certain forward flight speed. Therefore, with the propeller-induced downstream flowfield quantified, the changes in tandem back propeller effective angle of attack can be modeled using the linear superposition method.

C) PIV Results for Back Propeller Inflow in Tandem Configuration at $\theta = 90°$

Phase-locked PIV experiments are conducted to further investigate the inflow of the back propeller in the tandem configuration and compare it with the results from the standalone propeller in the previous section. The results for the back propeller inflow flowfield for the selected d_z configurations are shown in Figure 13 for $0.1 \leq J \leq 0.5$. The annotated propeller location represented the propeller disk location with all images taken at a 90-degree propeller phase angle.

At $\theta = 90°$, the propeller inflow angle (α_{in}) is equal to the freestream flow angle based on Equation 3. At $J = 0.1$, the propeller at $d_z = -0.4$ experience the lowest α_{in} as a result of the induced downwash of the front propeller as seen in Figure 11, the measured α_{in} for $d_z = 0.0$ and $d_z = 0.4$ are similar with the latter case showing a slightly higher α_{in}. A significant decrement in α_{in} for the $d_z = -0.4$ case is observed at $J = 0.2$

due to the direct impact of the front propeller wake, with almost no upwash due to the propeller tip vortex in the flowfield observed.

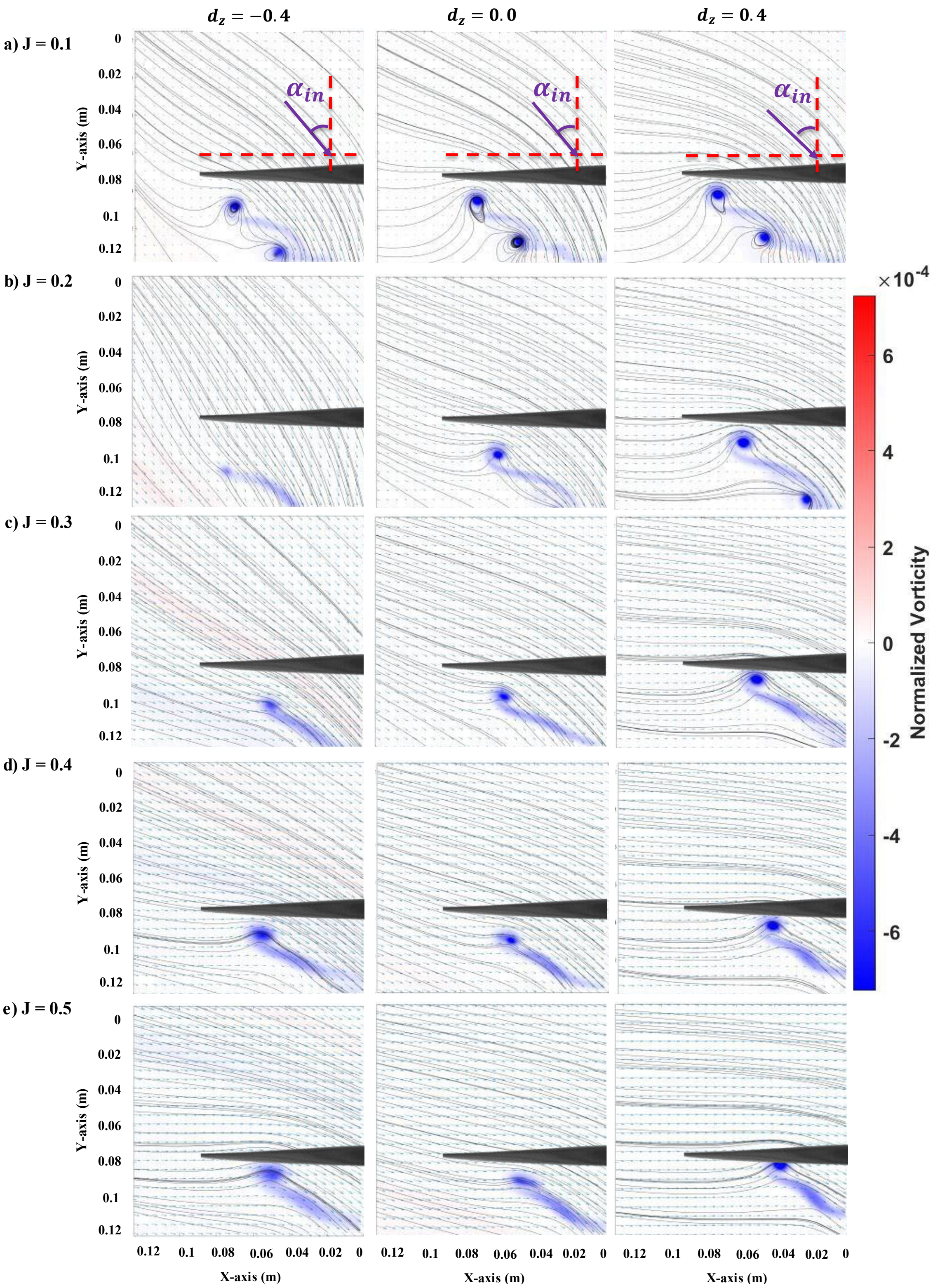

Figure 13. PIV Results for Selected Back Propeller Inflow at $\theta = 90°$ a) $J = 0.1$, b) $J = 0.2$ c) $J = 0.3$ d) d) $J = 0.4$ and e) $J = 0.5$

With a further increment in J, an increment in α_{in} is observed for the $d_z = -0.4$ case as the front propeller wake starts to move away from the back propeller disk. The difference in the inflow of $d_z = 0.0$ and $d_z = 0.4$ case also increases, with the latter one almost resembling the inflow distribution of the standalone propeller at $\theta = 90°$ (Figure 11). The overall trend is preserved with the increment of J, it is worth noting that the overall flowfield for the $d_z = -0.4$ cases become more and more similar to the $d_z = 0.0$ cases at $J \geq 0.5$. This agrees with the results from our previous study [21,22] where the overall propeller performance for these two cases is also similar at $J \geq 0.5$, indicating the similarity in the propeller performance when the propeller inflow matches.

Figure 14 shows the measured back propeller α_{in} distribution for all tandem propeller configurations at $d_y = 1.25$, along with the α_{in} distribution of the standalone propeller at $\theta = 60°, 75°$ and $90°$ (Figure 8) for comparison under various advanced ratios. Overall, the results are very similar to the induced inflow angle measurement based on the front propeller downstream flowfield in Figure 11. At $J = 0.1$, a higher d_z leads to a higher α_{in} distribution with the results from $d_z \geq 0.4$ seeing the highest α_{in} distribution which is almost identical to the standalone propeller at $\theta = 90°$, and the $d_z = -0.8$ case having the lowest α_{in} distribution, due to the direction front propeller wake interaction. It is also worth noting that the $d_z = 0.0$ case also closely resemble the result from $\theta = 75°$.

At $J = 0.2$, the overall trend and the magnitude order of α_{in} distribution follows the results in Figure 11, with α_{in} for the $d_z = -0.8$ recovers and $d_z = -0.6$ and $d_z = -0.4$ cases experiencing the lowest α_{in}. For the $d_z \geq 0.4$ cases, α_{in} distribution still matches the $\theta = 90°$ case at $r/R \leq 0.85$ as a lower upwash is measured closer to the tip of the propeller. This trend is clearer for the $d_z = 0.0$ case where the α_{in} distribution also matches the $\theta = 75°$ until it deviates closer to the blade tip with a much lower upwash measured.

The aforementioned trend remains preserved at higher J tested, where the $d_z \geq 0.4$ cases almost perform like the standalone propeller. This also has almost negligible performance penalty on the back propeller or the tandem rotor system. The $d_z = 0.0$ case experience an induced inflow angle that resembles the flight condition of a standalone propeller at $\theta = 75°$ and the $d_z < 0.0$ cases showing even lower α_{in}, and thus a higher performance penalty.

D) Linear Superposition of the Back Propeller Flowfield and Thrust Prediction

Combining the flowfield of the standalone propeller (Figure 8 and Figure 11) using the linear superposition method provides an estimation of the back propeller inflow angle in tandem configuration, which can be used to compare with the measured back propeller inflow angle in Figure 14. This is illustrated in Figure 15. Since the back propeller rotational speed varies, the u and v velocity component in the flowfield is normalized by the propeller tip speed using a similar idea as the propeller advance ratio, where:

$$J_u = \frac{u}{nD} \qquad J_v = \frac{v}{nD} \tag{5}$$

This is then used to calculate the estimated back propeller inflow angle, $\alpha_{in\ Est}$ using Equation 5.2 below:

$$\alpha_{in\ Est} = \operatorname{atan}\left(\frac{u_{Single\ Inflow} + u_{Single\ Wake} - V\infty}{v_{Single\ Inflow} + v_{Single\ Wake}}\right) = \operatorname{atan}\left(\frac{J_{u\ Single\ Inflow} + J_{u\ Single\ Wake} - J}{J_{Single\ Inflow\ SI} + J_{v\ Single\ Wake}}\right) \tag{6}$$

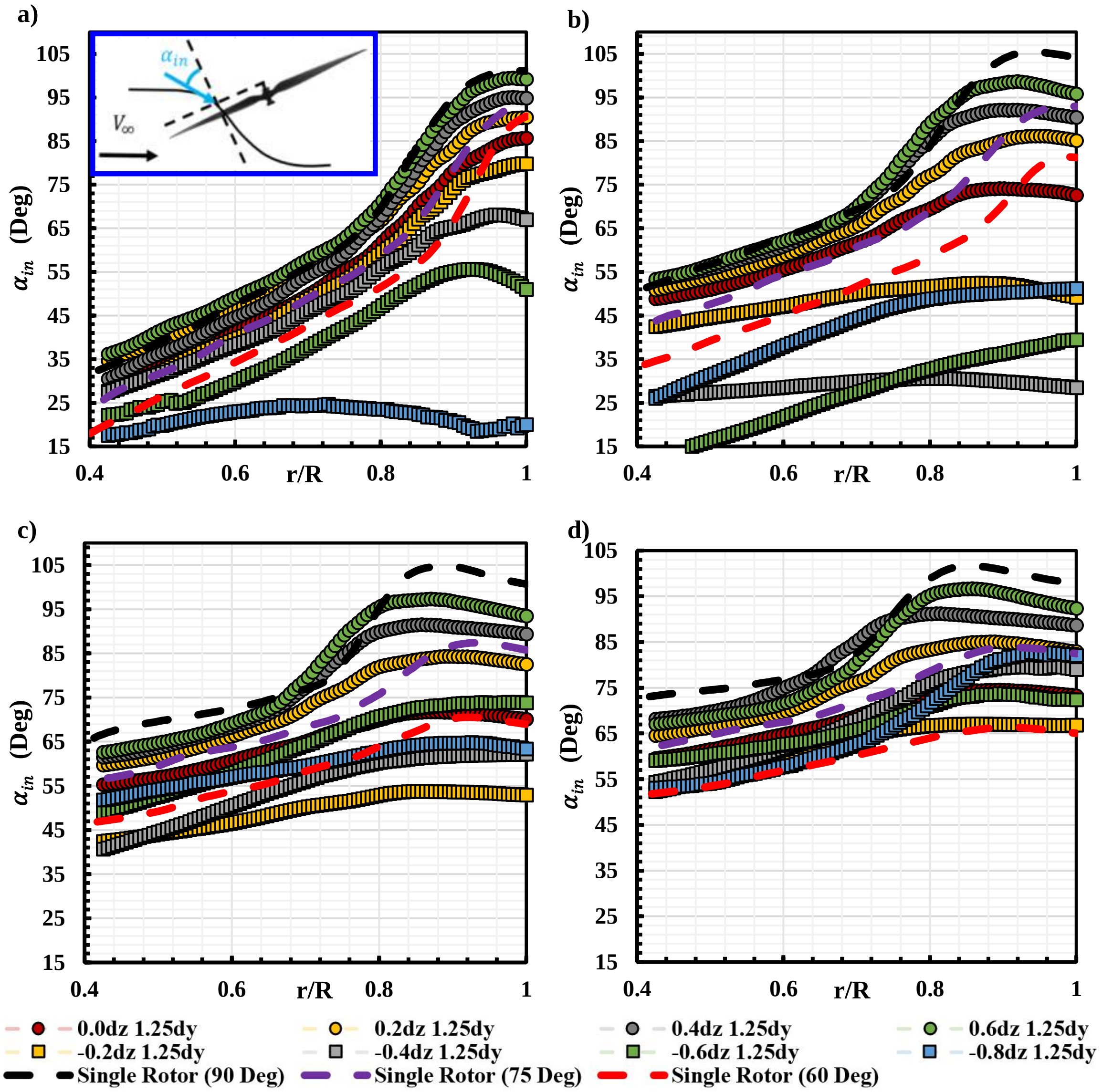

Figure 14. Measured Back Propeller Inflow Distribution in Tandem Configuration at a) $J = 0.1$ b) $J = 0.2$ c) $J = 0.3$ and d) $J = 0.4$.

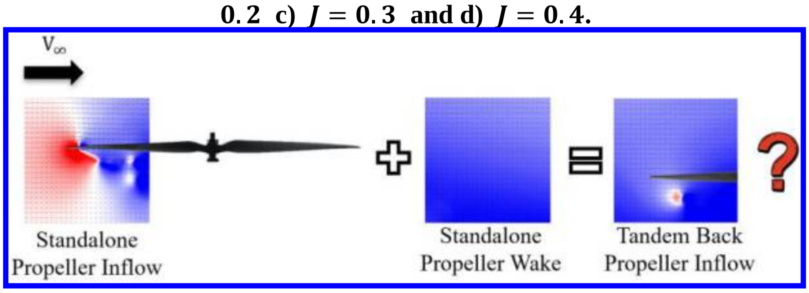

Figure 15. Applying Linear Superposition Method for Back Propeller Performance Prediction

Where the subscript 'Single Inflow' represents the normalized induced velocity component of the standalone propeller inflow and subscript 'Single Wake' represents the normalized induced velocity component downstream of the standalone propeller. As the u velocity component contains the propeller upstream freestream velocity, the freestream velocity component is then subtracted from the summation of the u velocity component.

The resulting estimated back propeller inflow angle, $\alpha_{in\,Est}$, for the negative staggered configurations are shown in Figure 16, and the positive staggered configurations are shown in Figure 17, along with the results for the measured α_{in} in Figure 14. For all configurations, $\alpha_{in\ Est}$ preserves the overall trend of the measured α_{in}. At $d_z \geq 0$, $\alpha_{in\ Est}$ underestimated the overall magnitude of α_{in} closer to the root of the propeller blade while overestimated the propeller upwash closer to the propeller tip, especially for small d_z cases, which is also observed for positive staggered configurations. Meanwhile at $d_z < 0$, $\alpha_{in\ Est}$ overestimate the magnitude of α_{in} at small advance ratios. At $J \geq 0.3$, a better agreement between $\alpha_{in\ Est}$ and α_{in} is observed. This trend can also be seen in Figure 18 when plotting the $\alpha_{in\ Est}$ and α_{in} againest d_z.

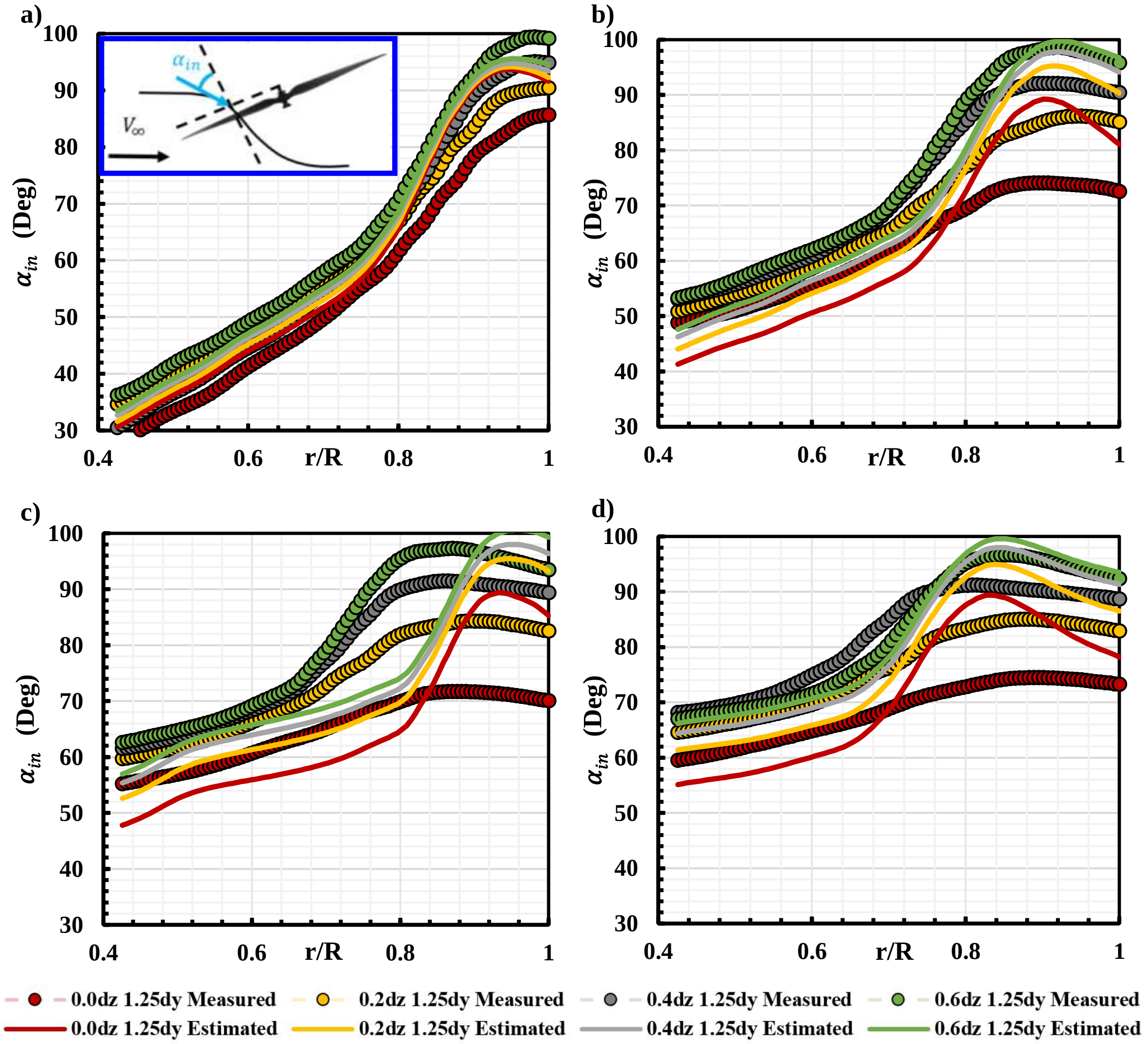

Figure 16. Measured and Estimated Back Propeller Inflow Distribution in Negatively Staggered Tandem Configuration at a) $J = 0.1$ b) $J = 0.2$ c) $J = 0.3$ and d) $J = 0.4$.

With the information of $\alpha_{in\ Est}$ at various J, the equivalent propeller incidence angle can be estimated using the experimental results from Figure 9 with a second-order surface interpolation. Then, the propeller incidence angle and advance ratio information are applied to the propeller prediction model developed in our previous study [26] to obtain an estimated propeller axial thrust coefficient. The results are shown in Figure 19, with a 4-degree α_{in} correct which is the difference in α_{in} at $d_z = 0.4$ at $J = 0.1$ in Figure 16. At $J = 0.1$, the estimated C_{TA} has a relatively good agreement with the experimental results as the difference between $\alpha_{in\ Est}$ and α_{in} is small. This difference increases between $0.2 \leq J \leq 0.4$ due to a higher difference between the $\alpha_{in\ Est}$ and α_{in}, and then reduces at higher forward flight speed. The average percentage error of the proposed prediction model is ~8.9%.

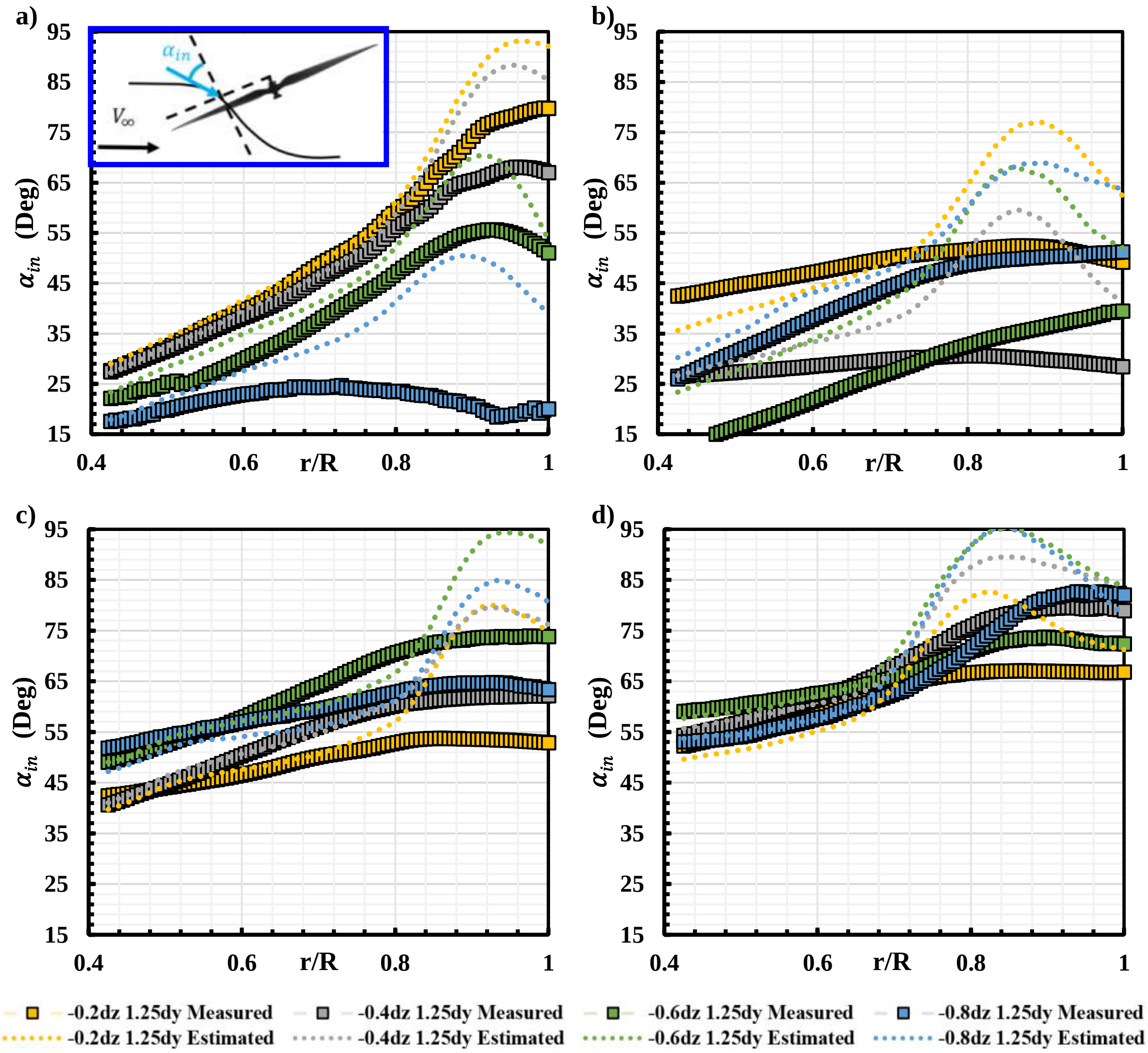

Figure 17. Measured and Estimated Back Propeller Inflow Distribution in Positively Staggered Tandem Configuration at a) $J = 0.1$ b) $J = 0.2$ c) $J = 0.3$ and d) $J = 0.4$.

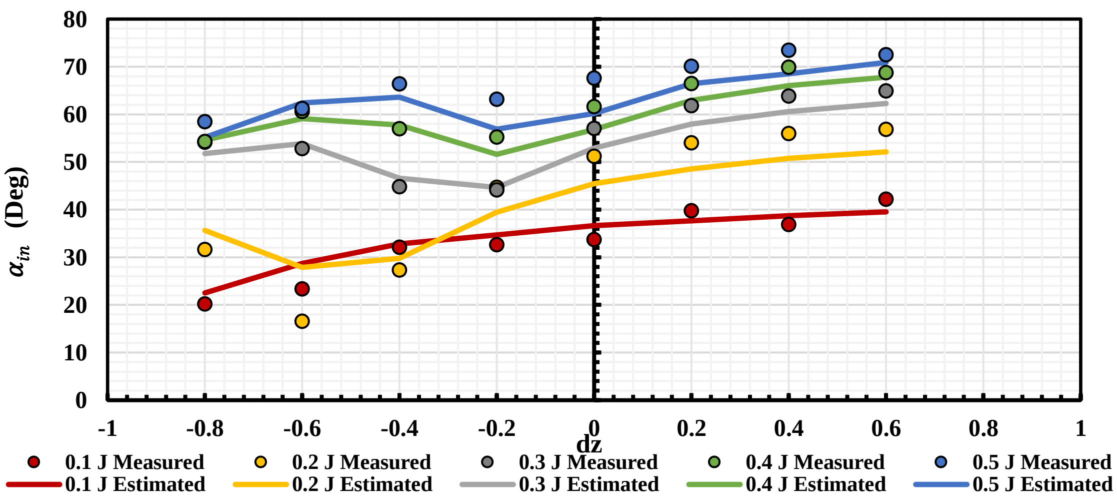

Figure 18. Measured and estimated α_{in} $vs. d_z$ att various advanced ratios

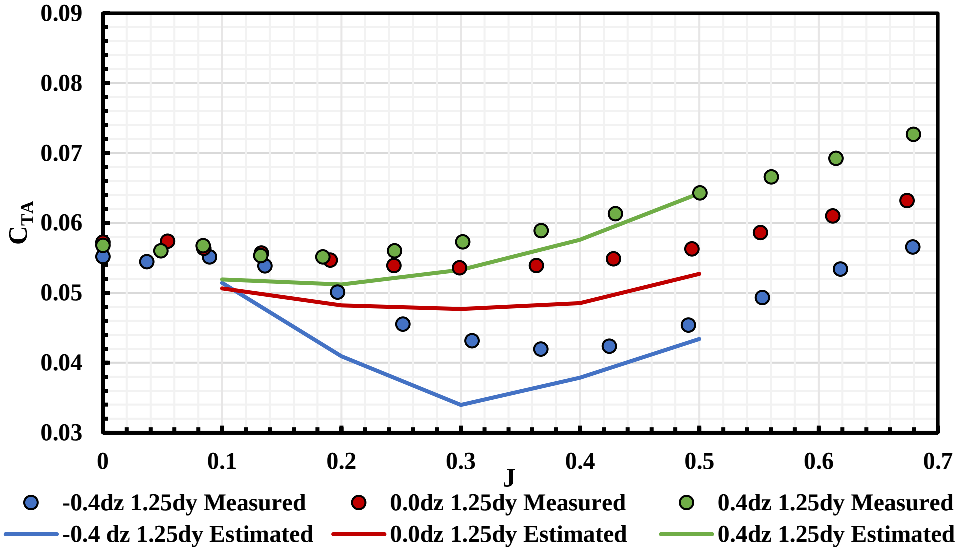

Figure 19. Measured and estimated back propeller C_{TA} $vs.J$ at selected d_z

V. Conclusions

In the current study, flow visualization was conducted to gain a basic understanding of the propeller flowfield in tandem configuration. Results indicate a reduction of the back propeller inflow angle when compared to the standalone propeller. This is then confirmed by the quantitative measurement from the PIV experiment. When increasing the negative stagger distance, a higher back propeller inflow angle is measured as a result of smaller propeller interaction. On the other hand, the changes in back propeller inflow angle are sensitive to both forward flight speed and negative stagger distance. A significant reduction of the back propeller inflow is observed when the front propeller wakes directly impacts the back propeller, leading to a reduction of the back propeller thrust generation and overall efficiency.

Utilizing the PIV measurement from the standalone propeller inflow and downstream induced flowfield, the linear superposition method is implemented to predict the changes in the back propeller performance in forward flight, by quantifying the changes in its inflow angle. Although the exact changes in the inflow angle do not match with the PIV measurement in the tandem configuration, the overall trend matches considerably well for both positive and negative staggered configurations. Therefore, using the flowfield information of a standalone propeller, the overall performance trend for a tandem staggered propeller configuration can be estimated at different flight conditions. This provides a guideline on the propeller placement when designing a multi-rotor vehicle with a tandem staggered propeller configuration.

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